# Experiment 07

## Title

Iris Flower Classification using Multiple Machine Learning Models

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| **SAP ID:** | 590013857 |
| **Dataset:** | `https://www.kaggle.com/datasets/uciml/iris/data` |

## AIM

To classify iris flower species using multiple machine learning models and compare their performance based on accuracy.

## THEORY

Iris classification is a supervised learning problem where the goal is to classify flowers into species based on their features.

Key concepts:

- **Classification Models**: Logistic Regression, Decision Tree, Random Forest, SVM, KNN
- **Feature Scaling**: Important for distance-based models like KNN and SVM
- **Confusion Matrix**: Shows classification performance per class
- **Accuracy**: Measures overall correctness of predictions

## Commands/Code Used

```
# Core steps

# Load dataset
df = pd.read_csv("Iris.csv")

# Preprocessing
# - Drop unnecessary columns
# - Label encoding
# - Train-test split
# - Feature scaling

# Models used
models = [
    LogisticRegression(),
    DecisionTreeClassifier(),
    RandomForestClassifier(),
    SVC(),
    KNeighborsClassifier()
```

```
]

# Evaluation
# Accuracy + Confusion Matrix
```

# Results

## 1. Confusion Matrices

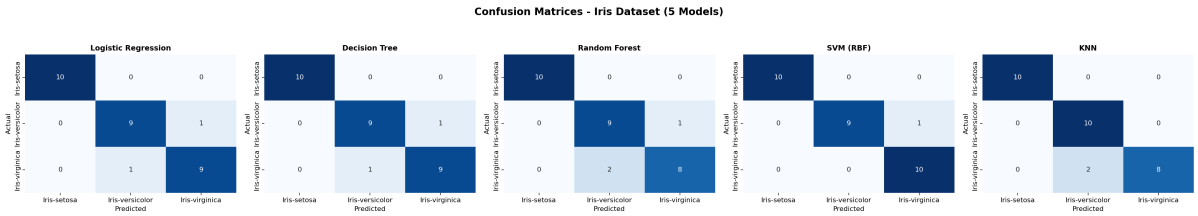


*Figure 1: Confusion matrices of classification models on Iris dataset*

## 2. Model Accuracy Comparison

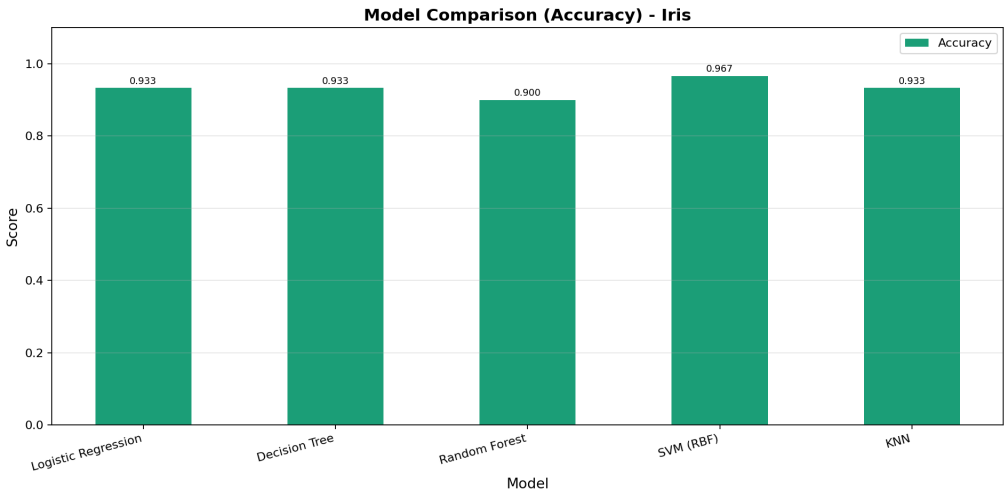


*Figure 2: Accuracy comparison of classification models*

# Conclusion

In this experiment, multiple classification models were applied to the Iris dataset. The results showed that SVM achieved the highest accuracy, followed closely by Logistic Regression, Decision Tree, and KNN.

Confusion matrices indicated that most misclassifications occurred between similar classes such as versicolor and virginica.

This experiment demonstrates the effectiveness of different machine learning algorithms in solving classification problems and highlights the importance of model comparison.